

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/369997729>

Gamification and the History of Art in Secondary Education: A Didactic Intervention

Article in Education Sciences · April 2023

DOI: 10.3390/educsci13040389

CITATIONS

5

READS

212

1 author:



[Mario Corrales Serrano](#)

Universidad de Extremadura

123 PUBLICATIONS 591 CITATIONS

SEE PROFILE

Article

Gamification and the History of Art in Secondary Education: A Didactic Intervention

Mario Corrales Serrano 

Departamento de Didáctica de las Ciencias Sociales, Lengua, y Literatura, Facultad de Formación del Profesorado, Universidad de Extremadura, 10003 Cáceres, Spain; mariocs@unex.es

Abstract: Within the curricula of social and civic competence that are taught in secondary education, Art History is a discipline that is usually relegated to a secondary position in terms of its role and the time dedicated to it. This report presents a didactic intervention for the systematic introduction of gamification strategies aimed at teaching Art History. Its objective is to analyze the performance of gamification strategies in the teaching of social sciences. The products generated in this intervention, applied longitudinally in three academic years, have been analyzed using qualitative methodology, with the WebQDA software. The results show a growth in motivation for the study of Art History and an uneven assessment of the different motivation strategies used in the process, although in general terms, a positive assessment of the gamification strategies used is presented.

Keywords: art history; gamification; longitudinal study; qualitative analysis; didactics; social sciences

1. Introduction

Art is the language in which civilizations and cultures express themselves in each of the periods of development in the history of humanity. Through various styles and forms, the artists of each era reflect their feelings, emotions, historical processes, human processes and traits of the identity of each moment in history [1]. The purpose of this report is to analyze the way in which the teaching-learning processes of Art History content are developed in the Secondary Education stage. Previous studies analyzed show how these contents very often occupy a secondary role, mainly due to the accumulation of subjects and the prioritization of other content [2,3].

To deal with this situation, the work carried out below explores the possibilities offered by gamification methodologies as motivating and revitalizing instruments for learning.

The first step of this research consists of a survey of the theoretical parameters that affect the teaching-learning process of Art History in Secondary Education, as well as the knowledge of tools that can be used to address the problems observed in this process [4]. At first, an analysis is carried out on the role that art contents play within the general context of Geography and History (In the Spanish educational system, the subject in which content and skills related to the social sciences are developed (geography, history and history of art) receives the generic name of Geography and History. The theoretical study and the didactic intervention that are presented have been developed in the context of this subject. Hence, this generic name is used hereinafter.) subjects at this stage, both at the level of importance and at the level of performance. Subsequently, an analysis of the incidence of gamification strategies in social sciences, and more specifically, in Art History, has been developed in order to assess the possibility of using this methodology to boost the teaching of these contents [5].

1.1. Teaching Art History in Secondary Education

The teaching of Art History is the tool through which students acquire the ability to obtain an idea about space and time in each period of history [2]. Within the various



Citation: Corrales Serrano, M.

Gamification and the History of Art in Secondary Education: A Didactic Intervention. *Educ. Sci.* **2023**, *13*, 389. <https://doi.org/10.3390/educsci13040389>

Academic Editor: James Albright

Received: 20 March 2023

Revised: 8 April 2023

Accepted: 10 April 2023

Published: 12 April 2023



Copyright: © 2023 by the author. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

contents associated with social and civic competence, Art History contents are of great value, to the extent that they contribute to the integral development of the person, both personally and for their professional future [6]. The competency model, which is intended to ensure that students finish their educational trajectory having acquired skills that allow them to develop their potential in life, discovers in the history of art a very important tool for the development of key elements of the person [7]. This importance is reflected in various studies on social and civic competence in the Compulsory Education Stage [3,8,9]. This is also expressed by Fernández, Fontal and Calaf in their work on this issue: “artistic language is essential for the development of sensitivity and a critical sense towards art and, in general, towards life” [10] (p. 5).

The fact of having knowledge about the different artistic manifestations that have occurred throughout history allows people to develop the ability to make an interpretation of social phenomena, to position themselves critically in front of them, to dialogue with the different cultures of the various periods of humanity and establish a connection with those factors that have determined the social processes of other times and with those of their own time [11]. Fernández and Muñoz highlight these contributions of artistic education:

“The contents on Art History can be presented as a valuable tool for the integral formation of the student, as a student, as a person and also as a citizen. Through these, the student has the possibility of accessing and learning a new language, developing their sensitivity and even forming a personal aesthetic judgment”. [12] (p. 231)

In the Compulsory Secondary Education stage (hereinafter ESO, as it is called in Spain), the teaching of content and skills related to the history of art are included within the various subjects that are designated by the generic name of Geography and History [13]. These subjects, which are taught with an unequal number of hours in the four courses of the compulsory stage, mix skills including physical geography, universal history and the history of Spain from the Prehistoric period to the Contemporary Age, political and economic geography, among others. The contents of Art History are integrated as part of this set, without their own specific hours, but rather as one more content of this block [11,14]. This stage is precisely the period in which students are put in communication with a huge number of styles, artists and works. This learning contributes to people’s ability to mature, understand and contextualize the world in which they live [15].

For this reason, it is necessary as a starting point for this work to make a reflection about the history of art that is taught to secondary school students. In this line, Batet [16] carries out an analysis about the value of the image as a resource for teaching social sciences and highlights the importance of using heritage-type resources to improve the acquisition of social and civic skills. Trepas [17] points out the importance of the presence of art in formal education, exploring various methodological possibilities.

Some of the contributions that are identified as important in the history of art are symbolic competence, the ability to develop the imagination, the identification and expression of emotions and the relationship between aesthetics and culture [18].

If we refer to those bases that must be taken into account for the didactic development of Art History, some authors [3,4,19] point out the limitations of the social sciences curriculum for art teaching and the need to promote didactic strategies and practical and active approaches to teaching these skills.

These limitations are evident, in turn, in the incidence that artistic content has in textbooks. The huge amount of material means that, on many occasions, the teaching of Art History occupies a secondary role, and this becomes relegated to the criteria of the teacher who teaches the subject. The relatively poor presence of this content in Geography and History manuals [20] is an indicator of this secondary role. Both the proportionality of content and the use of images and artistic representations in these manuals is a matter that generates debate in the scientific literature [21–23]. In general, it is obvious to consider that the presence of this kind of content in these manuals is limited, scarce and with little depth.

Other papers [19,24] analyze the causes of the secondary role occupied by the history of art, highlighting the excessive amount of curricular content and the prioritization of other types of content. In his study, he also highlights the importance that the current educational system seems to give to the knowledge and conservation of artistic heritage, although this is not later reflected in the development of competencies.

Another of the signs of this abandonment of art teaching can be seen in the data reported by student enrollments, both in the modality of arts studies and in the subject of Art History, which is an optional course in the 2nd year of a baccalaureate. Thus, as can be seen in the enrollment statistics for university entrance exams, only three percent opt for the arts modality and ten percent usually enroll in the optional subject of Art History [25]. This, at the same time, leads to difficulties for the development of the teaching of Art History at the university level [26,27]. Among the factors that affect these choices, we find the vision that the student has about the usefulness of knowledge or its possibilities for practical application.

From the arguments presented, it is concluded that on the one hand, the teaching of art is a fundamental competence for the integral development of the person and enhances the preference for the aesthetic, the symbolic vision and the development of multiple languages to express and be able to interpret emotions and feelings. On the other hand, it is concluded that the way in which these curricular contents are taught in the Secondary Education stage has a series of limitations linked to the lack of time, the excess of content in the subject in which it is taught and the prioritization of other content. This being the case, is it possible to improve the place that Art History occupies in the Secondary Education stage in the current educational system? What resources can be used for this purpose? How can we influence these teaching-learning processes?

1.2. Gamification as a Useful Teaching Strategy for Social Sciences

Among the various methodological strategies of an active type that have been introduced in the teaching-learning processes in recent years, gamification strategies present a series of advantages that can be used to introduce improvements in the teaching-learning process of social science content [28,29]. Although there is no consensus about a definition of this concept of educational innovation [30], talking about gamification is not talking about the use of games in education, but about a structuring of the teaching-learning process, introducing the rules of the games, as such, to influence motivation and connection with the process itself [31]. Thus, in the business field, for example, gamification is defined as follows:

“Gamification is a technique, a method and a strategy at the same time. It starts from the knowledge of the elements that make games attractive and identifies, within a specific activity, task or message, in a Non-game environment, those aspects that can be converted into a game or ludic dynamics. All this to achieve a special involvement, on the part of the users, encourage a change in their behavior or transmit a message or content. That is to say, create a significant and motivating experience”. [32] (p. 14)

This system, which was born in the business world and has been introduced in the educational field in the last decade, presents a high motivating power for students and also represents an interesting incentive to generate links between the student and the skills to be developed [33,34].

Among the most interesting qualities of this classroom strategy, the following can be highlighted [35]:

- Breaks with the traditional vision that students have about a typical class session, and introduces contexts close to the students themselves and their preferences.
- Transforms the role that teachers and students usually play in the classroom.
- Provides rules and regulations with which to articulate a student-learning connection process.
- Has a high motivational power [36].

Different previous experiences show the suitability of using this methodology for teaching social sciences. Rivero-Gracia presents a series of successful experiences of applying this methodology in the area of teaching [37]. Corrales analyzes the effects of using this methodology on the motivation of students in learning the subject of Geography and History [38]. Ordoñez Ocampo and his collaborators show that the use of this methodology in social sciences encourages creativity, reflection and critical argumentation, as well as the development of independence in the learning process [39]. Another interesting factor to support the use of this methodology in social sciences is the development of the emotional processes that it puts into operation and that are linked to learning, causing positive effects [40,41].

Taking into account its nature as an emerging active methodology, the motivational effects that it causes in students, the characteristics of its application to teaching-learning processes and existing experiences in the field of teaching social sciences, it is possible to consider gamification as a relevant methodology for the design of a strategy that improves the teaching of social sciences.

1.3. Possibilities of Application to the History of Art

The general context described in the application of gamification in the area of social sciences is reinforced in studies that present experiences on the incorporation of gamification for learning content linked to art and historical heritage. This is the case of the study by Jiménez-Palacios and López [41], which analyzes the possibility of using video games as a resource for learning about heritage in Huelva, with interesting results on an emotional level. In this same order of ideas, Corrales studies the effects of this methodology for knowledge about heritage and the museum in the city of Badajoz [42,43]. Fernández (2019) analyzes the effect on the roles of the teacher and the student with the application of this methodology in the teaching of art.

In the scientific literature, research of a very diverse nature is reviewed on this subject for specific contents, such as the teaching of Romanesque art [5], or for sections of the learning process, such as evaluation [44]. Likewise, some studies show the suitability of this methodology for learning STEAM disciplines, which include the teaching of art [45,46].

Everything said so far is useful as an endorsement for the design of a didactic intervention process that uses the gamification methodology as a strategy to improve the teaching of content and skills related to the history of art in the Secondary Education stage.

2. Materials and Methods

To carry out this research, a didactic intervention has been designed for the harmonious and systematic development of contents and skills related to the history of art in the Compulsory Secondary Education stage. Once the intervention was designed and applied, a qualitative analysis of the products generated by the intervention activities was carried out longitudinally over a period of 4 academic years. The results obtained in this analysis are reported in the corresponding section.

2.1. Objectives

The theoretical framework of this work has revealed some of the problematic pedagogical circumstances in which the teaching-learning processes of the competences linked to the history of art in Secondary Education are developed. Some of these circumstances, such as the curricular conditions or the distribution of the subject, are given to the teacher. However, there is a wide field of action from the field of active methodologies to be able to influence this process and introduce improvements. In this sense, we can appreciate how active learning methodologies, such as gamification, are shown to be effective solutions to demotivation or lack of interest that could be caused by the difficulties described. This can be observed in the aforementioned studies, not only at a general level, but also in the area of social sciences and in relation to learning about art.

From this context, the present study focuses on analyzing the results of the designed didactic intervention, which consists of the systematic application of gamification tools in the teaching of Art History. Its main objective is to assess the suitability of gamification methodologies to improve the teaching-learning processes of Art History skills in Secondary Education. It is based on the hypothesis that the application of gamification strategies has a positive impact on these teaching-learning processes, on student motivation and on learning results. If this hypothesis is verified, it can be affirmed that this didactic strategy contributes to improving the situation of learning Art History skills, despite the fact that some of the difficulties described do not depend on teacher performance nor can be easily modified in this performance.

In order to complement this main objective, a series of additional objectives are also taken into account that are related to learning about Art History skills, generating a positive vision of these contents in students and the different types of gamification methodology that have been applied throughout the entire intervention.

These secondary objectives are the following:

1. Analyze the academic and evaluation results of the contents and skills related to the history of art in relation to the rest of the subjects.
2. Analyze expressions and fragments that express a positive vision of art on the part of the students.
3. Assess the impact on student learning of the different gamification strategies, highlighting as far as possible their effect on the students in the sample.

The analysis procedures applied to the products generated in the various phases of the educational intervention have allowed the assessment of the degree of achievement for these objectives, as shown in the Section 3.

2.2. Participants

The sample on which the didactic intervention was applied was made up of a total of $n = 47$ students from the Secondary Education stage, divided into two groups/class (22 and 25 students upon reaching 4th ESO) who belonged to different study centers. Due to the longitudinal nature of the study, the intervention began to be applied in the 2017/2018 academic year, when the students were in the 1st year of ESO, and concluded in the 2020/2021 academic year, when the sample was in the 4th year of ESO. The very longitudinal nature of the study meant that, due to student failures or center transfers, it went from an initial sample of 52 students to the final figure of 47. Depending on the contents taught in the Geography and History subject, the 3rd year of Secondary Education did not have activities for the application of the intervention, since in its curricular contents there are no competences related to Art History. The selection of this sample was made at convenience, with a non-probabilistic nature, to be able to adapt to the requirements involved in carrying out a longitudinal study.

Regarding the characterization of the sample, it presented a gender distribution of 53% female and 47% male. In relation to the academic results, the overall average grade obtained in the subject of Geography and History in the courses in which the intervention was carried out was 6.13 (in the Section 3, these data are analyzed in more detail in comparison with the grades obtained in Art History content). The average age of the students in the sample was 12 years at the beginning of the intervention and 17 years at the end of it. In relation to their predisposition to work with gamification methodologies, none of these groups/classes had been subjected to systematic application interventions of this methodology, so they were not predisposed to it in any way. As additional information, in both groups there were students with special educational needs (2 in one group and 3 in the other).

2.3. Didactic Strategy

The didactic strategy that was used as the main structure for the intervention is a strategy with a combination of different gamification methods. These methods were interspersed in the various academic courses, depending on the characteristics of the

students in each course, the number of hours available for the application of the intervention and each method's suitability based on the skills and content to be developed in each subject of each course. The causes that justify the choice of this type of methodology are linked, on the one hand, to the high motivational power generated in the students by the application of strategies linked to games, as previously stated. On the other hand, it was taken into account the fact that this methodology energizes teaching and learning processes both individually and in groups.

Regarding the application of this methodology in the area of knowledge of social sciences, and specifically to the teaching and learning of artistic skills, there are different studies cited above that support its suitability. Despite this, a series of risks that were taken into account in the design of the intervention are described, such as the different way in which each student faces each gamification strategy, demotivation for not achieving objectives or excessive competitiveness.

The different methodological variants of gamification that have been applied in each of the courses are described below:

- Learning by discovery: In this learning model, students are allowed to take a leading role and increase their motivation around an objective that is set for them to discover through research strategies [47].
- Narrative gamification: This is a strategy that allows students to be introduced to an alternative context to that of the classroom context itself, introducing the learning process in a non-academic narrative [48]. In this way, there is an impact on the reversal of demotivation processes that are sometimes generated by the academic dynamics themselves.
- Gamification by obtaining individual and group rewards: With this type of strategy, the achievement of short learning objectives is encouraged, and the student is challenged to achieve new objectives [49,50]. The inclusion of group rewards helps to reduce the risk of excessive competition.
- Escape Room and Break Out: These types of strategies, applied at specific moments of the process, allow a high dose of motivation and also place students in front of new challenges for the application of learned skills [51–53].
- Gamified evaluation tools: Through this type of resources, some of the negative effects of the evaluation processes are transformed through the application of strategies that are typical of games, which also allow the better performance of the students in the evaluation [54].

2.4. Description of the Didactic Intervention

The design of the didactic intervention responds to the objective set by the research: that is, to assess the suitability of the gamification methodology as a useful tool to develop content and skills related to the history of art in the Secondary Education stage. For this purpose, a series of programmed activities were developed with various gamification strategies in each of the academic years in which content linked to the history of art is taught.

The design of the intervention and its subsequent analysis are longitudinal in nature. This type of intervention has the advantage of allowing the evolution of results to be observed over a determined period of time [55]. The analysis of results of this type makes it possible to assess the incidence of the continued use of the methodology with the same sample of students and at different stages of their career [56,57].

These activities have continuity between the various academic years through the gamification narrative, which connects some experiences with others and contextualizes them appropriately in the historical period studied in each academic year. The academic years in which the didactic intervention was developed were the 1st, 2nd and 4th of ESO.

The reasons for choosing these specific academic years were the following:

- These are the courses in which the contents of the different periods of the history of art are distributed.

- They belong to the compulsory schooling period, and the subject of Geography and History is mandatory for all the students of the stage in these courses. This allows us to assess the incidence of the intervention, not only in the students who choose Art History subjects (as happens in the 2nd year of baccalaureate education in Spain), but in the general population of students of the stage.
- The 3rd academic year of Secondary Education is excluded, since the curricular contents and skills of this course are focused on human and economic geography and do not address artistic content.

Table 1 shows the academic years in which this didactic intervention was developed and the specific contents of history of art that were worked on through it.

Table 1. Academic years in which the didactic intervention and curricular contents were developed. Self-made.

Academic Year	Art Period	Content
1st ESO	Prehistory	Appearance of the rites, material and artistic remains, painting and sculpture (p. 135).
	Ancient Age (Egypt and Mesopotamia)	Describe some architectural examples from Egypt and Mesopotamia (p. 135).
	Ancient Age (Classical period)	Identify and describe the characteristic features of Greek and Roman works of art, differentiating between those that are specific (p. 136).
2nd ESO	Middle Age	Describe characteristics of Romanesque, Gothic and Islamic art (p. 136).
2nd ESO	Modern Age	Identify features of the Renaissance and Humanism in European history starting from different types of historical sources Learn about the works and legacy of artists, humanists and scientists of the time (137). Identify significant works of Baroque art (p. 137).
4th ESO	Contemporary Age	Identify the main works and styles of the 19th century.

Source: Law Decree 98/2016.

The designed didactic sequence followed the same pattern in all academic years despite the fact that various gamification strategies were used. In this didactic pattern, the following elements were presented:

- A narrative framework from which students are encouraged to be curious about the knowledge of the history of art of the historical period studied in this course.
- The implementation of a process based on challenges, according to which the student has to achieve a series of objectives related to the narrative story that has been raised.
- The development of micro-projects in which the student acquires knowledge about periods, works of art and specific artists.
- Obtaining a series of rewards, which allow the student to advance within the gamified narrative, acquire content about the skills and enjoy motivational elements for the execution of the designed didactic sequence.
- The evaluation of the learning process through gamified mechanisms that allow verification of the knowledge acquired.
- The entry in the daily learning program for each of the micro-projects that are carried out in each intervention phase.

Figure 1 represents the fundamental elements of the didactic pattern used.

Regarding the specific development of the didactic intervention in each of the academic years, the distribution is as follows:

(a) 1st ESO: Explorers of the ancient art

In the first year of Secondary Education, the intervention is presented as a learning experience by discovery. The students of the sample are immersed, through the gamification narrative, in a team of explorers who have the mission of traveling back in time to document the great works of art that appear in the first era of human history. This documentation task makes it possible to reproduce these works of art in the future thanks to the information collected. The mission of the explorers of the art of the past is to describe these works as well as possible through a short presentation.



Figure 1. Fundamental elements of the didactic pattern of the intervention. Self-made.

- The starting point is an audiovisual material in which the gamified narrative is shown to the students: the students receive the proposal to become explorers of ancient art. They are given a passport that accredits them as such and allows them to travel through time.
- The curricular contents addressed are those of the art corresponding to Prehistory and the Ancient Age. In this context, knowledge about prehistoric works of art, such as megalithic monuments, cave paintings, ziggurats, pyramids, sphinxes, Greek temples, Greek sculpture and Roman monuments, is addressed.
- The gamification process consists of solving four micro-projects designed as exploration missions. In each of them, the students of the sample, divided into groups, must know about the work of art that is assigned to them, explain its characteristics, prepare a reproduction and make a presentation in the classroom using technological resources (QR, augmented reality, virtual reality ...). As a reward, the students receive a stamp in their passport, so that if they receive all the stamps, they can access the final evaluation game of the activity.
- As a final product of the entire experience, a series of materials are obtained to decorate the classroom, which also allows the approach to all these works of art by all of the students.
- Finally, all the students participate in a final evaluation game, designed with Kahoot, for the evaluation of contents related to these works of art.
- For the evaluation of the experience, each student opened a learning diary. After each of the missions, they must write an entry in their diary in which three elements appear: Periods of art that have been learned, characteristics of those periods and examples of works and the assessment of art.

(b) 2nd ESO: Virtual museum

In the 2nd year of ESO, the intervention is presented as a learning experience based on challenges through which they obtain rewards and progress towards a final challenge.

- The first step is a proposal that the students receive through a symbolic letter written to them by the great artists of the medieval and modern times. Given the experience acquired in the previous course, the students have been selected to build a virtual museum with works of art from the medieval and modern periods. In this way, it will be possible to maintain the memory of these artists in the current era.
- The curricular contents studied are those corresponding to Romanesque art, Islamic art, Gothic art, Renaissance and the Baroque period.
- Each group of students carries out a total of 5 micro-projects, analyzing a work of art in each of them. The work consists of making a digital document in which the image of the commented work of art is located alongside a short historical summary and an artistic commentary that includes the fundamental characteristics of said historical period.

- Overcoming each micro-project gives the student groups a qualification, the symbolic reception of a thank you letter from the artists who are the authors of the works they have studied and the possibility of using a class session to play learning games. Obtaining all the thank you letters allows students to complete a puzzle with which they access the final challenge via QR code: an evaluation game.
- As a final product, a virtual museum is obtained in which the students of the sample can visit all the works of art studied by each group and in each historical period. In total, 44 works of art from these periods were analyzed.
- The final evaluation challenge consists of carrying out a digital Break Out game, in which the students must use the knowledge acquired to solve each enigma and reach the end. At the end of the challenge, the students receive a diploma with a QR code that they can only read in the next phase of the intervention.
- For the assessment of the experience, each student makes an entry in the learning diary for each micro-project, detailing the elements that have already been described before.

(c) 4th of ESO: Monuments Mission

In the 4th year of ESO, the intervention is presented as a learning experience based on challenges. The starting point is the diploma with the QR code from the previous phase. Through this code, students access the viewing of the film “The Monuments Men” (The synopsis of the film is as follows: In the context of World War II, Lieutenant Frank Stokes (George Clooney) commands a platoon of art experts with a mission: to recover from the clutches of a German army that is marching towards the debacle as many looted works of art as possible. The work of Stokes and his men will therefore be the only hope to recover this priceless heritage.). Supported by the narrative of this film, the students are placed on the scene of their participation as a squad of artistic special forces for the recovery of works of art from modern and contemporary times.

- The first step is the viewing of the film, after which they are commissioned to symbolically become that squad to recover the memory of those works of art. The way to recover the memory of these works is to carry out research micro-projects on them.
- The curricular contents addressed are those of art corresponding to the modern and contemporary periods.
- The gamification process consists of preparing classroom presentations with content about the works of art of that time. Each group makes a total of three presentations throughout the academic year. The reward obtained for the presentation of each work is a piece of a puzzle. By reconstructing the puzzle, the students discover that this is the invitation to participate in the final Escape Room of the activity.
- As a final product, the students make an audiovisual set with all the rescued works of art and their respective comments.
- The evaluation procedure consists of carrying out an Escape Room session set in the Second World War, in which they need to use the art contents they have learned to be able to leave the classroom.
- With each work on which they do an activity, the students make an entry in the learning diary, following the same parameters as the previous times. Figure 2 summarizes the didactic intervention and the gamification process.

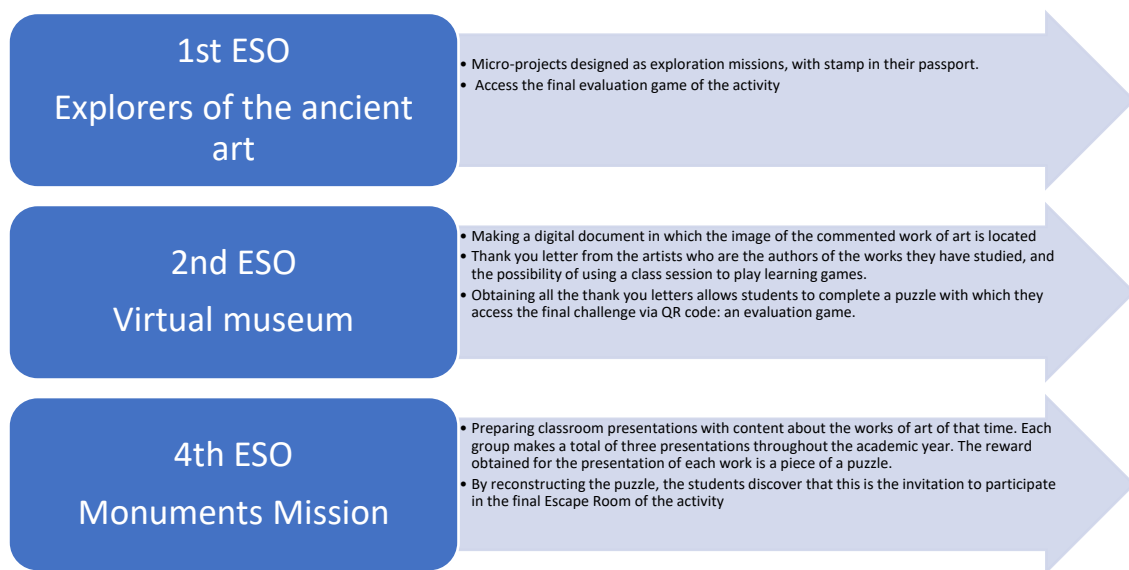


Figure 2. Structure of the gamification process by courses. Self-made.

Table 2 presents the distribution of gamification strategies in each application phase.

Table 2. Distribution of gamification strategies in each phase of the intervention. Self-made.

Academic Year	1st ESO	2nd ESO	4thESO
Discovery learning	Yes	No	No
Challenge-based learning	No	Yes	Yes
Narrative gamification	Yes	Yes	Yes
Use of rewards	Yes	Yes	Yes
Escape room	No	No	Yes
Break Out	No	Yes	No
Evaluation games	Yes	Yes	Yes

2.5. Qualitative Research Sources

The didactic intervention applied in the previously described conditions has generated a series of products that are useful as sources of analysis in order to obtain information concerning the objectives of this research. Figure 3 shows the structure that the students in the sample followed in their entries in their learning diaries.

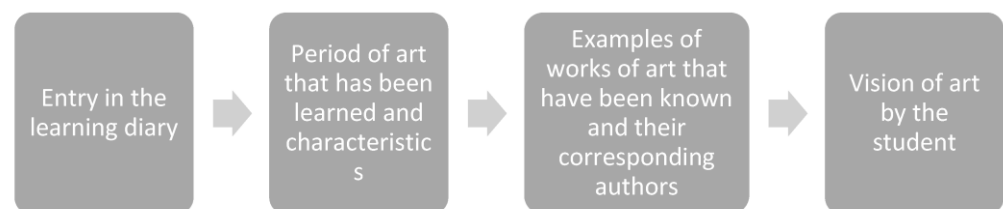


Figure 3. Structure of entries in the learning diary proposed to students. Self-made.

Throughout the three research phases, a total of 536 textual sources with the structure described in the figure were obtained. Those sources that did not respond to the designed structure have been eliminated. This structure has allowed subsequent analysis based on the research objectives, showing interesting results that are presented in the corresponding section.

2.6. Instrument and Investigation Procedure

The analysis of these sources has been carried out from a qualitative approach, since they are textual sources. In relation to the different moments in which these sources were obtained throughout the entire process, this study was of a longitudinal nature since information related to the research objectives was obtained at different moments throughout four academic years. This longitudinal character is shown in Figure 4, which details the moments of obtaining the texts that serve as a source for the research.

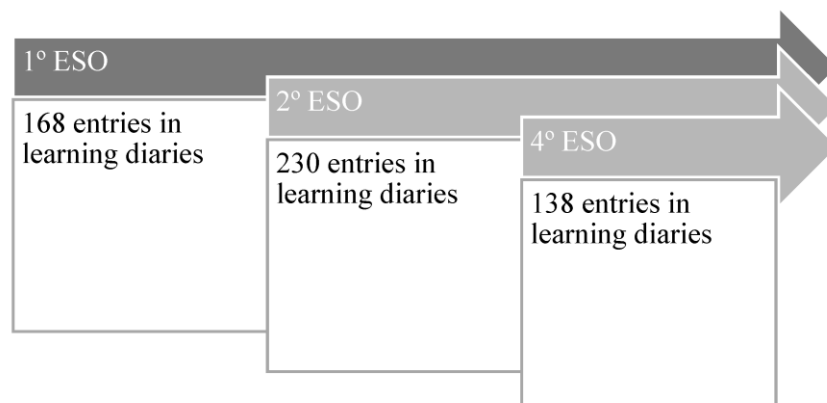


Figure 4. Number of sources depending on each phase of intervention. Self-made.

To carry out this analysis, the qualitative analysis software WebQDA was used. This software allows for the categorization of non-numeric sources in order to obtain results linked to their content. This type of analysis shows valuable results in the field of social science studies [58]. Based on the objectives of the study, a categorization structure of the sources was designed in which three main aspects were analyzed: learning results, vision observed in students in the sample in relation to the contents and competencies about the history of art and assessment of the gamification strategies used in the development of the didactic intervention. In the analysis, a gradual categorization of these aspects has been taken into account, as well as the evolution of the vision that the students were presented with in the different stages of the study.

3. Results

The analysis procedure of the sources obtained in the didactic intervention allowed the obtaining of results in relation to the research objectives that were raised at the beginning of this work. These results related to such objectives are shown below, based on the parameters of the qualitative analysis that has been carried out.

3.1. Learning Results

In relation to the learning of skills related to the history of art, and to compare with the academic results generally obtained by students in the subject of Geography and History, two ways of measuring were used:

On the one hand, the average grades obtained in the general assessment tests of the subject in each academic year in which the intervention is applied are compared with the results shown by the gamified assessment tools on content about art. Figure 5 shows this comparison (The global grades of “Geography and History” correspond to the curricular contents of this subject in each course in the Spanish educational system. In the different courses, geography and history contents are mixed, separated by didactic units.).

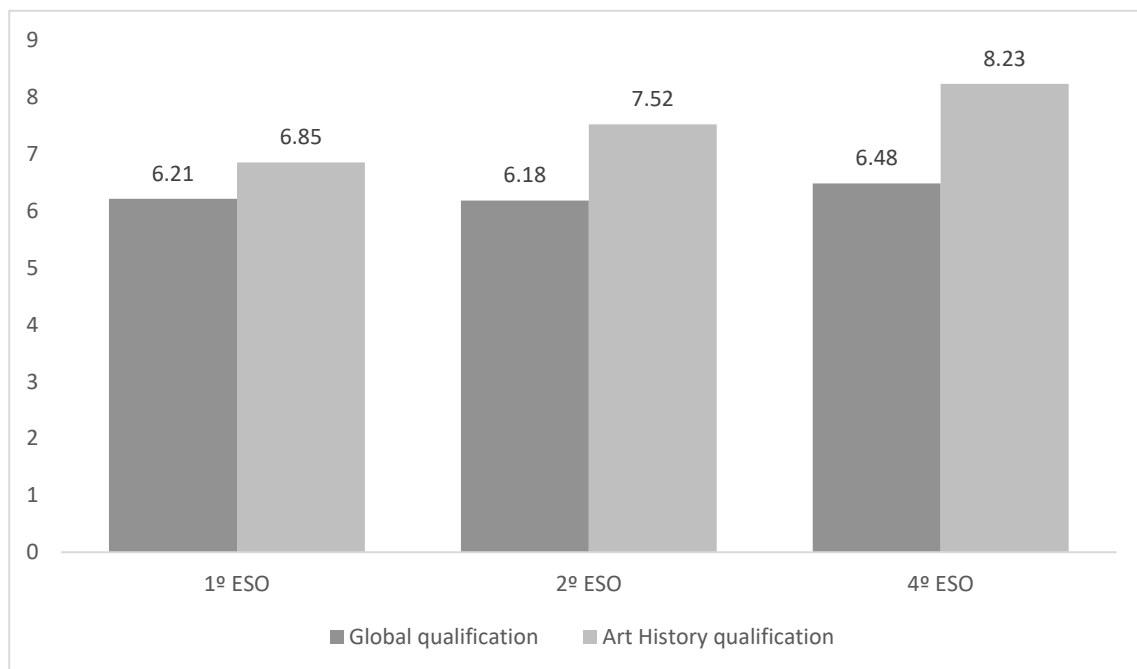


Figure 5. Comparison of average grades of the academic years with the grades obtained through the gamified assessment tools for the Art History content. Self-made.

These results show two interesting issues: the general grades obtained in the subject have remained stable, while those obtained with the gamified instruments in the Art History content have progressively increased. In addition, in all the courses better grades are obtained for the Art History content than the general average. Based on these data, it can be concluded that there is a positive effect on the learning of Art History skills in the students participating in the intervention.

On the other hand, when analyzing the textual sources in which the students narrate their learning, a progressive improvement in learning about the contents can be seen, as shown in Table 3.

Table 3. Learning outcomes narrated in the sources. Self-made.

	Memorize Artistic Styles	Memorize Authors	Memorize Works of Art
1st ESO	52%	36%	72%
2nd ESO	62%	86%	82%
4th ESO	67%	52%	79%

These results allow us to positively assess the incidence of the intervention in the learning of contents and historical-artistic competences. Additionally, a longitudinal evolution is observed according to which, as the intervention phases progress, the knowledge that the students have in relation to the analyzed parameters improves.

3.2. Vision about the History of Art by the Students

In relation to the vision that the students show about the contents of Art History, the sources have been categorized in relation to the items shown in Table 4. It can be observed how the explicit or very explicit presence of expressions in which the subjects show a positive vision about the history of art predominates.

Table 4. Results of the assessment of the history of art by the students as narrated in the sources. Self-made.

Intervention Phase	(a) No References Detected	(b) Indirect References Are Detected	(c) Explicit References Are Detected	(d) Very Explicit and Abundant Presence
1st ESO	26	36	70	26
2nd ESO	19	27	79	55
4th ESO	15	25	93	58
Total	60	88	242	139
Percentage	11.5%	16.5%	44.7%	27.3%

The following texts, extracted from the sources, show examples of classification of each of the categories collected to make an interpretation of the degree of assessment that the students reflect in relation to the contents of Art History (No examples of categorization with option (a), since the searched references are not detected in these fragments):

Fragments categorized with option (b):

“Thanks to this mission I have known Romanesque art. It has caught my attention, because it is very different from other types of art that I knew, and it has been curious to see how some Romanesque buildings are still in the middle of the countryside”.

“I had never really understood the painting of 20th century painters. It seemed to me that it didn’t make any sense. Getting to know these works through the activity we have carried out has helped me to relate it to everything that is happening in history, and now I see more sense in it”.

Fragments categorized with option (c):

“It has been very interesting for me to know the types of Greek sculpture and see how it has evolved in a short time. Before this activity I thought that sculpture was more or less the same at all times”.

“I knew some of the paintings by Renaissance artists that I have been studying, but I did not know what they were called or who they belonged to. Thanks to this activity I have known their authors, their characteristics, and I have been able to look for more paintings by Leonardo, which I really liked”.

Fragments categorized with option (d):

“Learning contents of works of art is what I like the most about the subject in that course. I have really enjoyed the challenge of getting to know works of art from the Renaissance, which are my favorites”.

“After all the activities that we have carried out in this course and in the previous ones, I think that art is one of the things that I liked the most about the subject, and I hope to continue learning things about art in Baccalaureate2”.

These results reveal a positive vision that was generated progressively in the students of the sample as progress was made in the various phases of implementation of the longitudinal intervention, which allowed us to consider the second research objective as achieved.

3.3. Assessment of the Different Gamification Strategies

In relation to the gamification strategies, as perceived by the students, the sources analyzed show favorable and unfavorable references, so that effects that positively and negatively affected the objective pursued with this study can be observed.

Table 5 shows these results.

Table 5. Assessment of students on gamified strategies. Self-made.

Academic Year	1st ESO		2nd ESO		4th ESO	
	Fav	Unfav	Fav	Unfav	Fav	Unfav
Discovery learning	68%	32%	-	-	-	-
Narrative Gamification	71%	29%	65%	35%	81%	19%
Use of Rewards	91%	9%	88%	12%	72%	28%
Escape room	-	-	-	-	66%	34%
Break Out	-	-	92%	8%	-	-
Assessment games	77%	23%	-	-	-	-

In the following fragments of the sources are presented some examples of the texts categorized according to each element analyzed.

Favorable assessment of the gamified narrative: “One of the fun parts of this activity is that you get into a story, and you hardly even realize that you are doing class work. You get into the role and try to achieve what you have been asked to achieve. That’s more fun than, memorizing names and dates”.

Unfavorable evaluation of evaluation games: “The games that we have done at the end, like Kahoot etc . . . are very stressful. I have become very nervous and I have failed some questions that I knew”.

Favorable evaluation of evaluation games: “I have had a lot of fun with the evaluation games, although it gave me a little overwhelmed when I saw that my teammates went up on the scoreboard and I did not. It’s a fun way to see what you know”.

Favorable evaluation of rewards: “That of having rewards with each mission that you beat is very good, because it makes you want the next one to arrive soon, and you are curious to know what the next reward will be”.

Unfavorable evaluation of Escape Room: “During the Escape Room I got very nervous. At first I wanted to do it, but just looking at how time progressed and I couldn’t pass the test I got a bit blocked”.

Favorable evaluation of Escape Room: “To finish all the tasks it has been great to carry out an activity as fun as an Escape Room. When my group has managed to reach the end before the set time, it has been very gratifying”.

Having seen this analysis, it can be stated that in general, all the methodologies are highly valued. However, this statement can be qualified with the fact that some activities, such as the Break Out or Escape Room, can generate feelings of being overwhelmed, stressed or shutdown in a part of the students in the sample. This detail must be taken into account for the development of the intervention in subsequent applications.

4. Discussions and Conclusions

The results obtained through the qualitative analysis that has been developed allow us to contrast the conclusions with the information reported in the scientific literature about the application of gamification strategies in the learning of Art History skills.

Regarding the acquisition of skills in the area in which the intervention and gamification strategies have been applied, this study presents results of a positive impact of gamification on student learning. Some studies report benefits in learning. Zepeda and collaborators [49] review how the application of this type of methodology reports a positive attitude and results in the learning processes. Montoya Lemus [59] analyzes the incidence of these didactic strategies in the teaching-learning process on the social sciences. The results show significant improvements attributable to the application of gamification. Bakhanova [60] highlights the impact on a more participatory performance of students in the learning process thanks to gamification. Fernández-Arias and her collaborators [61] specifically analyze the suitability of gamification to acquire skills in the social sciences, studying specific processes and comparing these with specific skills in the social sciences. Their conclusion is solid:

“In this way, based on the results shown in this article, the result of a process structured in several stages to establish links between the different elements of gamification and social skills, the potential of gamification as a tool for any subject to acquire and develop necessary and demanded social competencies in today’s society”. (p. 404)

Asiri [62] goes one step further in this direction, concluding how the teacher can influence the attitude, perception and results of the students through the use of this methodology. All these studies are consistent with the conclusions obtained in this article, highlighting the positive effect of gamification on learning.

Regarding the positive incidence of gamification on the students’ vision of art and heritage, Autor [43] concludes in his study on the gamification of a museum visit that this technique improves the perception that students have about art and heritage. Escalera-Fernández [18] highlights the role played by the history of art in the introduction of gamification for the teaching of social competencies and concludes in his study that gamification favors a positive vision of art by students. Casado [5] exposes the same conclusion after a didactic experience of teaching Romanesque art.

Finally, in relation to the effects of the various gamification strategies, Segura and Parra [63,64] present positive results for the teaching of physical education through the Escape Room, although they also review the risks of generating contradictory emotions, such as reviewed in our work. In this same line, Sánchez-Martín [65] presents concerns that are expressed about the evaluation games. Positive evaluations are generally reported, as happens in the present investigation [32,44,66]. Reward dynamics are probably the most studied process in gamification. Research indicates satisfactory results of this methodology and a good impact on motivation, as can also be seen in our study [67]. Finally, it is also worth noting the effectiveness of the gamified narrative, revealed in studies such as that of [48].

The development of the research that has been carried out allows us to highlight two issues as a final reflection: on the one hand, this study verifies the suitability of the various gamification methodologies as a useful tool for revitalizing the teaching process and for learning content related to the history of art. Both the results obtained in all the phases of the intervention, as well as the fact that as the longitudinal application of these strategies progressed, improvements were detected in the academic performance and in the learning of the students, allow us to obtain this conclusion. On the other hand, it highlights the fact that learning methodologies have the capacity to improve structural problems in curricular design and in the development of the contents of the subject and to give a further point of interest to this.

Among the limitations of this work, it is worth highlighting the fact that this didactic intervention has only been applied in this study, so that these results cannot be compared with those obtained in other applications of the same. The inability to reorganize the content distribution of the Social Sciences subject is also detected as a limitation. The difficulty is structural, and to a large extent it does not depend on the intervention of the teacher, so it will therefore be necessary to replicate this application to consolidate the results obtained through this first application.

Funding: This research received no external funding.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: Data is unavailable due to privacy.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Artola, J.P. *Modelos y Teorías de La Historia Del Arte*; Universidad de Deusto: Bilbao, Spain, 2015.
2. Freedman, K. La Enseñanza Del Tiempo y Del Espacio: Comprensión de La Historia Del Arte y de La Herencia Artística. *Rev. Educ.* **1992**, *298*, 81–88.

3. Fuentes-Moreno, C.; Gil-Duran, N. La Didáctica de La Historia Del Arte y La Educación Competencial. Propuesta de Replanteamiento Curricular. *Enseñanza Cienc. Soc.* **2019**, *18*, 63–75. [CrossRef]
4. Sasot, M.L.R.; Martínez, M.G. Qué Historia Del Arte Enseñar En La ESO? *Iber* **2021**, *104*, 7–13.
5. Casado Martel, V. *El Arte Románico a Través de La Gamificación: Una Propuesta de Mejora Para Educación Secundaria*; Universidad de Almería: Almería, Spain, 2018.
6. Rocha-Cáceres, R. El Modelo Educativo Basado En Competencias Para La Enseñanza Del Arte. *Educere* **2016**, *20*, 215–224. Available online: <http://www.redalyc.org/articulo.oa?id=35649692003> (accessed on 22 September 2022).
7. Clark, P.; Sears, A. *The Arts and the Teaching of History*; MacMillan: New York, NY, USA, 2020; Available online: <https://www.palgrave.com/us/book/9783030515126#reviews> (accessed on 19 September 2022).
8. Rojas, R. El Patrimonio Cultural Andaluz Como Elemento Contribuyente a La Competencia Social y Cívica. *Redes Educ.* **2012**, *2012*, 1–8.
9. Ortega-Sánchez, D.; Pagès Blanch, J. The End-Purpose of Teaching History and the Curricular Inclusion of Social Problems from the Perspective of Primary Education Trainee Teachers. *Soc. Sci.* **2020**, *9*, 9. [CrossRef]
10. Villalvilla, A.F.; Masachs, R.C.; Merillas, O.F. Aprender Arte En La Escuela: El Humanities Project Guía La Experiencia Con Escolares de 8-10 Años. In *Arte Para Todos: Miradas Para Enseñar y Aprender el Patrimonio*; Ediciones Trea, S.L.: Trea, Spain, 2003; pp. 19–48.
11. López-Castelló, R. La Historia Del Arte Escolar En El Recuerdo Del Profesorado de Educación Secundaria En Formación. *Rev. Complut. Educ.* **2021**, *32*, 55. [CrossRef]
12. Fernández, J.M.; Muñoz, M.I.V. Qué Aprenden Los Alumnos de ESO Sobre Historia Del Arte. Un Análisis de La Legislación y Los Exámenes En El Caso de La Región de Murcia. *Educ. Siglo XXI* **2017**, *35*, 229–254. [CrossRef]
13. Prats, J. *Didáctica de la Geografía y la Historia*; Ministerio de Educación Cultura y Deporte: Madrid, Spain, 2011.
14. Trepát, C.A.; Rivero, P. *Didáctica de La Historia y Multimedia Expositiva*; Editorial Graó: Barcelona, Spain, 2010.
15. López, J.P.A. La Necesidad de Educar En Artes Visuales. *Arte Y Mov.* **2018**, *18*, 17–41. Available online: <https://revistaselectronicas.ujaen.es/index.php/artymov/article/view/3914> (accessed on 8 November 2022).
16. Batet, I.S. El Valor Del Arte y La Renovación de La Didáctica En Las Ciencias Sociales. *Iber* **1996**, 7–16.
17. Trepát Carbonell, C.-A.; Feliu Torruella, M. Propuesta En Formación En Didáctica de La Historia Del Arte: Aprender a Enseñar Historia Del Arte En La Secundaria. *Iber* **2010**, *8*, 86–90.
18. Fernández, I.E. Gamificación En La Educación: El Rol de La Historia Del Arte. *CIVINEDU 2019* **2019**, *2019*, 65.
19. Ávila Ruiz, R.M. La Enseñanza Del Arte En La Escuela. *Íber* **2003**, *37*, 30–42.
20. Gómez-Carrasco, C.J.; Puche, S.M.; Grau, B.P. Los Manuales de Ciencias Sociales y La Enseñanza de La Historia Del Arte En 2º de ESO. *Ensayos* **2012**, *27*, 69–88.
21. Guillot, H.R. Los Rostros de La Historia. La Enseñanza de La Historia Moderna En Los Manuales de Secundaria a Través de Sus Imágenes. *Enseñanza Cienc. Soc.* **2017**, *16*, 15–25. Available online: <https://raco.cat/index.php/EnsenanzaCS/article/view/334514> (accessed on 16 September 2022).
22. Rausell Guillot, H.; Talavera Ortega, M. *Género y Didácticas*; Tirant Humanidades: Valencia, Spain, 2019.
23. Valls Montés, R. Las Imágenes En Los Manuales Escolares de Historia y Las Dificultades de Su Uso Didáctico. *Clío Asoc.* **2007**, *11*, 11–23. Available online: http://www.memoria.fahce.unlp.edu.ar/art_revistas/pr.10351/pr.10351.pdf (accessed on 26 September 2022). [CrossRef]
24. Ruiz, R.M.Á. El Papel de La Historia Del Arte En El Currículum. *Iber* **2001**, *29*, 67–80.
25. Corrales-Serano, M.; Modeno-Losada, J.; Sánchez-Martín, J.; Zamora-Polo, F. Estudio cualitativo de las motivaciones del alumnado de Bachillerato en referencia a la modalidad de estudios. *Campo Abierto Rev. Educ.* **2020**, *39*, 85–99. [CrossRef]
26. Corrales Serrano, M.; Sánchez-Martín, J.; Moreno Losada, J.; Zamora Polo, F. The Role of the Social Sciences When Choosing University Studies: Motivations in Life Stories. *Educ. Sci.* **2021**, *11*, 420. [CrossRef]
27. Menéndez, M.D.C.R.; López, S.T.; García, C.M.F. El Impacto Del Género En Las Elecciones Académicas de Los Estudiantes Asturianos Que Finalizan La ESO. *REOP* **2006**, *17*, 239–260. Available online: <http://www.redalyc.org/articulo.oa?id=338230775007> (accessed on 18 September 2022). [CrossRef]
28. Delgado-Algarra, E.J. Gamification and Game-Based Learning: Motivating Social Sciences Education. In *Research Anthology on Developments in Gamification and Game-Based Learning*; IGI Global: Hershey, PA, USA, 2022; pp. 932–956.
29. Kim, E.; Rothrock, L.; Freivalds, A. An Empirical Study on the Impact of Lab Gamification on Engineering Students' Satisfaction and Learning. *Int. J. Eng. Educ.* **2018**, *34*, 201–216. Available online: <http://www.scopus.com/inward/citedby.url?scp=85042539814&partnerID=8YFLogxK> (accessed on 16 September 2022).
30. Huotari, K.; Hamari, J. A Definition for Gamification: Anchoring Gamification in the Service Marketing Literature. *Electron. Mark.* **2017**, *27*, 21–31. [CrossRef]
31. Gil-Quintana, J.; Prieto Jurado, E. La Realidad de La Gamificación En Educación Primaria. Estudio Multicaso de Centros Educativos Españoles. *Perf. Educ.* **2020**, *42*, 107–123. [CrossRef]
32. Marín, I.; Hierro, E. *Gamificación: El Poder Del Juego En La Gestión Empresarial y La Conexión Con Los Clientes*; Empresa Activa: Barcelona, Spain, 2013.
33. Gómez-Trigueros, I. Methodologies Gamified as Didactic Resources for Social Sciences. *IJET* **2019**, *14*, 193–207. [CrossRef]

34. Trigueros Ramos, R.; Navarro Gómez, N. La Influencia Del Docente Sobre La Motivación, Las Estrategias de Aprendizaje, Pensamiento Crítico y Rendimiento Académico de Los Estudiantes de Secundaria En El Área de Educación Física. *PSYE* **2019**, *11*, 137–150. [CrossRef]
35. Sailer, M.; Homner, L. The Gamification of Learning: A Meta-Analysis. *Educ. Psychol. Rev.* **2020**, *32*, 77–112. Available online: <https://link.springer.com/article/10.1007/s10648-019-09498-w> (accessed on 21 September 2022). [CrossRef]
36. Ardila-Muñoz, J.Y. Supuestos Teóricos Para La Gamificación de La Educación Superior. *Magis* **2019**, *12*, 84. [CrossRef]
37. Rivero-Gracia, M.P. Procesos de Gamificación En El Aula de Ciencias Sociales. *Iber* **2017**, *86*, 4–6.
38. Corrales Serrano, M. Gamificación Como Herramienta Para Educar En La Participación. *Clio* **2021**, 23–48. [CrossRef]
39. Ordoñez Ocampo, B.P.; Ochoa Romero, M.E.; Erráz Alvarado, J.L.; León González, J.L.; Espinoza Freire, E.E. Consideraciones Sobre Aula Invertida y Gamificación En El Área de Ciencias Sociales. *Rev. Univ. Soc.* **2021**, *13*, 497–504.
40. Corrales-Serrano, M. Emociones de Estudiantes Preuniversitarios En Ciencias Sociales Con Experiencias de Gamificación. *Investig. Esc.* **2020**, 84–96. [CrossRef]
41. Jiménez-Palacios, R.; López, J.M.C. La Enseñanza y Aprendizaje de Las Ciencias Sociales a Través Del Patrimonio, Videojuegos y Emociones. Estudio de Caso En Un IES de Huelva (España). *Panta Rei* **2021**, *15*, 103–133. [CrossRef]
42. Corrales-Serrano, M.; Sánchez-Martín, J.; Moreno-Losada, J.; Zamora-Polo, F. GIS in the History Classroom: Displaying the Walls of Elvas and Badajoz through the Use of Google Earth. *Proceedings* **2019**, *38*, 2. [CrossRef]
43. Corrales-Serrano, M.; Garrido-Valverde, J. Uso Del Patrimonio y Aprendizaje de La Historia de La Ciudad. Una Experiencia de Gamificación En La Alcreazaba de Badajoz. *Rev. UNES* **2020**, 30–40. [CrossRef]
44. Rodríguez, F.; Loro, F.; Villén, S. Experiencia de Gamificación En Alumnos de Magisterio Para La Evaluación de La Asignatura Sociología de La Educación Mediante El Uso de La Plataforma Kahoot. In Proceedings of the 3rd International Conference on Innovation, Documentation and Teaching Technologies, Kuala Lumpur, Malaysia, 17–18 January 2015; p. 223.
45. Parra-González, M.E.; López-Belmonte, J.; Segura-Robles, A.; Moreno-Guerrero, A.-J. Gamification and Flipped Learning and Their Influence on Aspects Related to the Teaching-Learning Process. *Heliyon* **2021**, *7*, e06254. [CrossRef] [PubMed]
46. López, P.; Rodrigues-Silva, J.; Alsina, Á. Brazilian and Spanish Mathematics Teachers' Predispositions towards Gamification in STEAM Education. *Educ. Sci.* **2021**, *11*, 618. [CrossRef]
47. De Miguel-González, R.D. Aprendizaje Por Descubrimiento, Enseñanza Activa y Geoinformación: Hacia Una Didáctica de La Geografía Innovadora. *Did. Geo.* **2014**, *14*, 17–36. Available online: <https://didactica geografica.age-geografia.es/index.php/didactica geografica/article/view/230> (accessed on 22 September 2022).
48. Barreal, J.; Jannes, G. La Narrativa Como Herramienta Docente Dentro de La Gamificación de La Estadística En El Grado En Turismo. *Dig. Educ. Rev.* **2019**, *36*, 152–170. [CrossRef]
49. Hernández, S.Z.; Mena, R.A.; Ornelas, E.L. Integración de Gamificación y Aprendizaje Activo En El Aula. *Ra Ximhai* **2016**, *12*, 315–325. Available online: <http://ilitia.cua.uam.mx:8080/jspui/handle/123456789/305> (accessed on 20 December 2022). [CrossRef]
50. Jeong, J.S.; González-Gómez, D.; Yllana Prieto, F. Sustainable and Flipped STEM Education: Formative Assessment Online Interface for Observing Pre-Service Teachers' Performance and Motivation. *Educ. Sci.* **2020**, *10*, 283. [CrossRef]
51. Ayuso Tapia, G. Creación de Un Escape Room Educativo Para Educación Secundaria Obligatoria. 2019. Available online: <http://uvadoc.uva.es/handle/10324/38983> (accessed on 22 September 2022).
52. Negre, C. BreakoutEdu. Microgamificación y Aprendizaje Significativo. 2017. Available online: <https://www.educaweb.com/noticia/2017/07/26/breakoutedu-microgamificacion-aprendizaje-significativo-15068/> (accessed on 21 September 2022).
53. Corrales Serrano, M.; Sánchez-Martín, J.; Moreno Losada, J.; Zamora Polo, F. Educar en el interés social de las ciencias sociales: Una intervención didáctica. *Rev. Prism. Soc.* **2021**, *35*, 249–276. Available online: <https://revistaprismasocial.es/article/view/4357> (accessed on 26 September 2022).
54. Suelves, D.M.; Esteve, M.I.V.; Chacón, J.P.; Marí, M.L. *Gamificación En La Evaluación Del Aprendizaje: Valoración Del Uso de Kahoot! In Innovative strategies for Higher Education in Spain*; Adaya Press: Eindhoven, The Netherlands, 2018; Volume 8. [CrossRef]
55. Delgado Rodríguez, M.; Llorca Díaz, J. Estudios Longitudinales: Concepto y Particularidades. *Rev. Española Salud Pública* **2004**, *78*, 141–148. Available online: <http://www.redalyc.org/articulo.oa?id=17078202> (accessed on 11 September 2022).
56. Arnau, J.; Bono, R. Estudios Longitudinales de Medidas Repetidas: Modelos de Diseño y Análisis. *Escr. Psicol.* **2008**, *2*, 32–41. Available online: <http://www.redalyc.org/articulo.oa?id=271020194005> (accessed on 21 September 2022). [CrossRef]
57. Salinero, J.G. Estudios Descriptivos. *Nure Investigación* **2004**, *7*, 1–3.
58. Carril-Merino, M.T.; Andreu-Mediero, B.; De la Calle-Carracedo, M.; López-Torres, E. Fuentes Orales Para El Desarrollo de La Empatía Histórica: Un Estudio En La Formación Del Profesorado de Educación Infantil. *Panta Rei* **2020**, *14*, 155–184. [CrossRef]
59. Montoya Lemus, D.A. Gamificación Como Estrategia Didáctica En Las Ciencias Sociales (Monografía de Compilación). 2021. Available online: <http://repository.uan.edu.co:8080/handle/123456789/4925> (accessed on 22 September 2022).
60. Bakhanova, E.; Garcia, J.A.; Raffe, W.L.; Voinov, A. Targeting Social Learning and Engagement: What Serious Games and Gamification Can Offer to Participatory Modeling. *Environ. Model. Softw.* **2020**, *134*, 104846. [CrossRef]
61. Arias, P.F.; Olmedo, E.O.; Rodríguez, D.V.; Vallecillo, A.I.G. La Gamificación Como Técnica de Adquisición de Competencias Sociales. *Prism. Soc. Rev. Investig. Soc.* **2020**, *31*, 388–409. Available online: <https://revistaprismasocial.es/article/view/3698> (accessed on 19 April 2022).

62. Asiri, M.J. Do Teachers Attitudes, Perception of Usefulness, and Perceived Social Influences Predict Their Behavioral Intentions to Use Gamification in EFL Classrooms? Evidence from the Middle East. *Int. J. Educ. Pract.* **2019**, *7*, 112–122. [[CrossRef](#)]
63. Parra; Segura, A. Análisis de Las Experiencias Gamificadas de Docentes y Alumnos de Educación Secundaria. *Espacios* **2019**, *40*, 15.
64. Segura-Robles, A.; Parra González, M.E. How to Implement Active Methodologies in Physical Education: Escape Room. *Educ. Sport Health Phy. Acti.* **2019**, *3*, 295–306. Available online: <http://hdl.handle.net/10481/56426> (accessed on 22 September 2022).
65. Zamora-Polo, F.; Corrales-Serrano, M.; Sánchez-Martín, J.; Espejo-Antúnez, L. Nonscientific University Students Training in General Science Using an Active-Learning Merged Pedagogy: Gamification in a Flipped Classroom. *Educ. Sci.* **2019**, *9*, 297. [[CrossRef](#)]
66. Gómez-Carrasco, C.-J.; Monteagudo-Fernández, J.; Moreno-Vera, J.-R.; Sainz-Gómez, M. Effects of a Gamification and Flipped-Classroom Program for Teachers in Training on Motivation and Learning Perception. *Educ. Sci.* **2019**, *9*, 299. [[CrossRef](#)]
67. Carrión-Candel, E. El Uso de La Gamificación y Los Recursos Digitales En El Aprendizaje de Las Ciencias Sociales En La Educación Superior. *DIM* **2018**, *36*, 7. Available online: <https://raco.cat/index.php/DIM/article/view/340828> (accessed on 22 January 2022).

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.